

Chemistry

1 The insoluble product formed is Barium sulphate $[BaSO_4]$ and color is white
 $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
(aq) (aq) (s) (aq)

2 Ammonia gas

3 Both assertion & reason are true. Reason is correct explanation of Assertion.

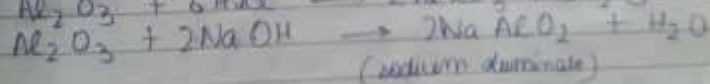
4 Assertion is true but reason is false.

5 The amphoteric oxides ~~are~~ are those which react with both acids as well as base in order to form salt and water.

Amphoteric oxides behave like

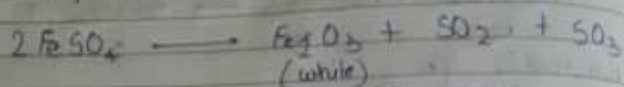
Examples: Zinc oxide (ZnO), Aluminium oxide (Al_2O_3)

Balanced equation:



6 The student will observe displacement reaction in beakers having Aluminium and Zinc strips.

7 On heating green vitriol crystals ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) following reaction is thermal decomposition takes place

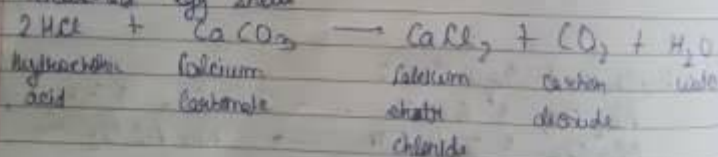


The colour changes from light green to white and due to formation of SO_2 and SO_3 pungent smells come.

8 (a) Zinc granules



(b) Crushed egg shells



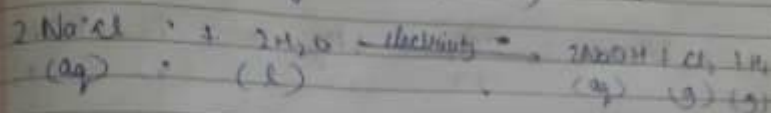
9 The salt used is Baking soda (NaHCO_3)



Baking soda is produced when CO_2 is passed from saturated solution of sodium carbonate.

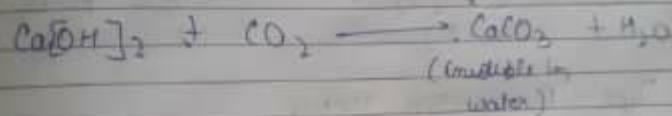
Uses = ① Soda-acid fire extinguisher
② To make baking powder and to make antacids

Q (a) Chemical equation involved in preparation of sodium hydroxide (NaOH).

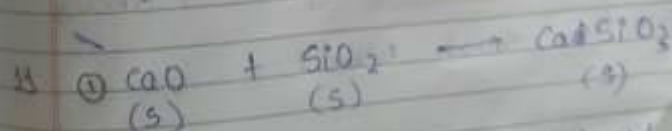
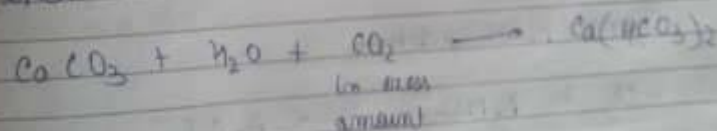


The process is called chlor-alkali process.

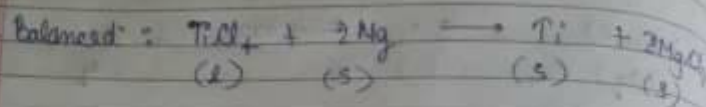
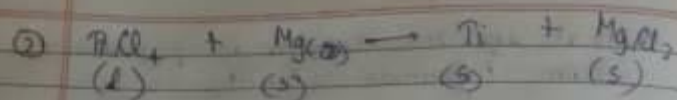
(b) When CO_2 is passed through lime water $[\text{Ca}(\text{OH})_2]$ it turns milky due to formation of calcium carbonate (CaCO_3).



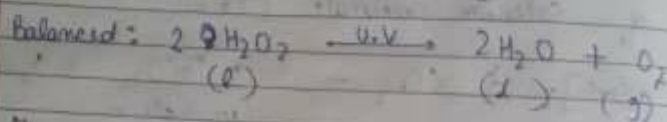
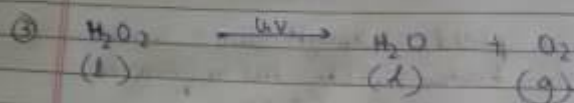
When excess CO_2 is passed, milky appearance due to formation of calcium bicarbonate $[\text{Ca}(\text{HCO}_3)_2]$ which is colourless & soluble in water.



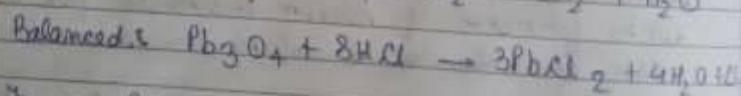
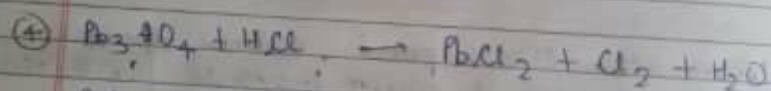
This equation is already balanced.
Type of chemical reaction: combination reaction.



Type: Displacement reaction



Type: Decomposition reaction



Type: Redox reaction



Balanced: It is already balanced.

Type: Double displacement reaction or precipitation reaction